Description of data pre-processing

2022-12-11

1 Introduction

The aim of this documentation is to illustrate how the data pre-processing is done for the data-set of the HP407 group project.

To replicate this study, you will need to install R, which is available at https://r-project.org/. For this preprocessing workflow, the current version of R on my computer is as follows:

```
R.Version()$version.string
```

```
## [1] "R version 4.2.2 (2022-10-31)"
```

To proceed, you must install and load the required packages, which is as follows:

```
# installation
install.packages("readxl")
install.packages("openxlsx")
install.packages("naniar")
install.packages("dplyr")

# loading
library(readxl)
library(openxlsx)
library(naniar)
library(dplyr)
```

2 Loading the dataset

Use the following code to load the dataset:

```
# load the dataset
dataset <- read_xlsx("Dataset.xlsx")
```

```
## New names:
## * `Intervention Intervention details` -> `Intervention Intervention
##   details...97`
## * `Intervention Intervention details` -> `Intervention Intervention
##   details...98`
## * `ADL after treatment SD` -> `ADL after treatment SD...352`
## * `ADL after treatment mean` -> `ADL after treatment mean...353`
## * `ADL after treatment N` -> `ADL after treatment N...354`
## * `ADL Baseline SD` -> `ADL Baseline SD...362`
## * `ADL Baseline mean` -> `ADL Baseline mean...363`
## * `ADL Baseline N` -> `ADL Baseline N...364`
## * `ADL after treatment mean` -> `ADL after treatment mean...391`
## * `ADL after treatment N` -> `ADL after treatment N...392`
## * `ADL after treatment SD` -> `ADL after treatment SD...393`
```

```
## * `ADL Baseline mean` -> `ADL Baseline mean...394`
## * `ADL Baseline N` -> `ADL Baseline N...395`
## * `ADL Baseline SD` -> `ADL Baseline SD...396`
## * `` -> `...5226`
```

```r
# replace the string value "-" with NA
dataset[dataset=="-"]  <- NA

# remove the rows all data item is NA
dataset <- dataset[rowSums(is.na(dataset)) != ncol(dataset), ]

# remove the columns all data item is NA
dataset <- dataset[,colSums(is.na(dataset))<nrow(dataset)]
```

When loading the dataset, R will alter some of the row names in case of incompatibility. After loading the dataset, we will first have a look at its size, with the following code:

```r
# show the size of the dataset
dim(dataset)
```

```
## [1]  733 3705
```

It is clear that this is a very large dataset with 733 rows and 3705 columns. This means the dataset contains 733 studies. We can first have a look at the first five rows of the dataset, with the following code to know what the columns might be about:

```r
# show first five rows
head(dataset)
```

```
## # A tibble: 6 x 3,705
##   Study ~1  Year Inter~2 Inter~3 Inter~4 Country Setting Types~5 Type ~6 Updat~7
##   <chr>    <dbl> <chr>   <chr>   <chr>   <chr>   <chr>   <chr>   <chr>   <chr>
## 1 Moraes ~  2008 Donepe~ Pharma~ Establ~ Brazil  partic~ Alzhei~ Dement~ Mild t~
## 2 Moraes ~  2008 Placebo No int~ No int~ Brazil  partic~ Alzhei~ Dement~ Mild t~
## 3 Liu 200~  2008 electr~ Tradit~ Acupun~ China   Hospit~ Vascul~ Dement~ Any se~
## 4 Liu 200~  2008 medica~ Pharma~ Non-es~ China   Hospit~ Vascul~ Dement~ Any se~
## 5 Liu 200~  2008 Scalp-~ Tradit~ Acupun~ China   hospit~ Vascul~ Dement~ Any se~
## 6 Liu 200~  2008 Wester~ Pharma~ Establ~ China   hospit~ Vascul~ Dement~ Any se~
## # ... with 3,695 more variables:
## #   `Is this a study focusing on carers of people living with dementia or mild cognitive impairment?
## #   `Primary Cognition Measure standardized z-score change` <chr>, Type <chr>,
## #   `Baseline Score - Primary Cognition` <chr>,
## #   `Number of Participants at Baseline` <chr>, `Baseline SD` <chr>,
## #   `Post Treatment Score` <chr>, `Post Treatment N` <chr>,
## #   `Post Treatment SD` <chr>, ...
```

The 3705 columns are not all useful to us, except for the first several columns containing basic information. Because it will be very difficult for us to read all the columns with either Excel or R, we can first output all these names to a seperate file with the following code:

```r
sink("colnames.txt")
colnames(dataset)
sink()
```

3 Screening by intervention type

Here we aim to filter the studies by the intervention type. First we can have a look at the column names related to "Intervention":

```
df_cols <- colnames(dataset)
df_cols[grep('Intervention', df_cols)]
```

```
##  [1] "Intervention"
##  [2] "Intervention types"
##  [3] "Intervention subgroups"
##  [4] "Intervention Brief description of intervention"
##  [5] "Intervention Brief description ofi ntervention"
##  [6] "Intervention Brief intervention description"
##  [7] "Intervention Brief introduction to the intervention"
##  [8] "Intervention Intervention details...97"
##  [9] "Intervention Intervention details...98"
## [10] "Intervention Intervention details (duration and intensity of the intervention, dosage of drugs
## [11] "Intervention intervention type"
## [12] "Intervention Intervention type.1"
## [13] "Intervention Interv. Type:"
## [14] "Intervention Interv. type [SELECT: Pharmaceutical; Traditional Chinese Medicine; Other trad. me
## [15] "Intervention Interv. type [SELECT: Pharmaceutical; Traditional Chinese Medicine; Other trad. me
## [16] "Intervention Name of intervention"
## [17] "Zarit Burden Interview (ZBI) Post Intervention Change mean"
## [18] "Zarit Burden Interview (ZBI) Post Intervention Change N"
## [19] "Zarit Burden Interview (ZBI) Post Intervention Change SD"
```

Among these variables, "Intervention types" seems to provide most relevant information for fast screening:

```
knitr::kable(table(dataset['Intervention types']))
```

Intervention.types	Freq
Comprehensive care interventions	7
Electrical brain stimulation	14
Multicomponent interventions	28
Music interventions	8
No intervention	251
Nutrition	4
Other interventions	2
Pharmaceutical	130
Physical exercise	17
Structured therapeutic psychosocial interventions	38
Supplements	44
Support for carers	14
Supportive psychosocial interventions	8
Traditional Chinese Medicine	156
Training and education for carers	6
Treatment of co-morbidities or additional risks	4
Unstructured therapeutic interventions	2

Thus, we will select 'Support for carers' and 'Training and education for carers'. But in case of any missingness, we also explore the studies under the intervention types of 'Structured therapeutic psychosocial interventions', 'Comprehensive care interventions', 'Supportive psychosocial interventions', 'Treatment of co-morbidities or additional risks', 'Unstructured therapeutic interventions' and 'Multicomponent interventions', but there is none related to support or education for caregivers.

```
df_studies <- dataset[dataset$'Intervention types' %in% c(
    "Structured therapeutic psychosocial interventions",
    "Training and education for carers",
    "Other interventions",
    "Support for carers",
    "Comprehensive care interventions",
    "Supportive psychosocial interventons",
    "Treatment of co-morbidities or additional risks",
    "Unstructured therapeutic interventions",
    "Multicomponent interventions"), ]
dim(df_studies)
```

```
## [1]  101 3705
```

Thus, we reduce the scope of the analysis from 733 studies to only 101 of them. Then we screen the further details of the intervention.

```
knitr::kable(table(df_studies$'Intervention subgroups'))
```

Var1	Freq
Cognitive stimulation therapy	2
Cognitive training	14
Combination of structured psychosocial interventions	3
Community-based interventions	1
Group support and therapy	7
Individual-based interventions	6
Information, education, and training to support carers	7
Multicomponent electrical brain stimulation + therapeutic substance	2
Multicomponent interventions	2
Multicomponent pharmaceutical + nutrition	1
Multicomponent pharmaceutical + psychosocial	1
Multicomponent pharmaceutical + TCM	7
Multicomponent physical exercise + psychosocial intervention	4
Multicomponent TCM	7
Multicomponent TCM + psychosocial or nursing intervention	4
Occupational therapy	3
Other interventions	2
Other structured therapeutic psychosocial interventions	5
Rehabilitation	4
Reminiscence therapy	7
Training and education for professional carers	2
Training and education for unpaid carers	4
Treatment of co-morbidities or additional risks	4
Unstructured therapeutic interventions	2

We surely do not want to include any drug-related intervention, so we exclude the terms including "TCM", "pharmaceutical", "nutrition", "therapeutic substance".

```
included_terms <- unique(df_studies$'Intervention subgroups')[- grep(
  "TCM|pharmaceutical|nutrition|therapeutic substance",
  unique(df_studies$'Intervention subgroups'))]
df_studies <- df_studies[df_studies$'Intervention subgroups' %in% included_terms, ]
```

```
df_studies <- df_studies[,colSums(is.na(df_studies))<nrow(df_studies)]
dim(df_studies)
```

```
## [1]   79 1244
```

Thus, we reduce the number of studies from 101 to 79. Then I output the information of the remaining studies to a excel file for further manual screening.

```
write.xlsx(df_studies, "Dataset_intervention_screening.xlsx")
```

The manual screening will check the details of the intervention, especially whether it involves support or training for carers. Thus, the following studies are selected:

```
selected_studies <- read.csv('selected_studies.txt')
knitr::kable(selected_studies)
```

Intervention.types	Study.Identifier
Comprehensive care interventions	Wang 2010b
Comprehensive care interventions	ZHAO 2010
Comprehensive care interventions	SUN 2010
Comprehensive care interventions	SU 2012
Comprehensive care interventions	JIANG 2012
Comprehensive care interventions	Wang 2014a
Comprehensive care interventions	Yang 2017a
Structured therapeutic psychosocial interventions	He 2012
Structured therapeutic psychosocial interventions	Novelli 2018
Structured therapeutic psychosocial interventions	Lok 2019
Support for carers	Pahlavanzadeh 2010
Support for carers	Wang 2012e
Support for carers	Aboulafia-Brakha 2014
Support for carers	Mercedeh 2014
Support for carers	Kamkhagi 2015
Support for carers	Söylemez 2016
Support for carers	Shata 2017
Support for carers	Salamizadeh 2017
Support for carers	Mahdavi 2017
Support for carers	Lök 2017
Support for carers	Orawan 2018
Training and education for carers	Amit 2008
Training and education for carers	TAN 2010
Training and education for carers	Guerra 2011
Training and education for carers	Wang 2017e
Training and education for carers	Wang 2017c
Training and education for carers	Lin 2017

4 Screening by column names

However, it will be hard in term of human labour to look thourgh the 3000+ columns for data extraction, so I use a series of keywords to screen the relevant columns for meta-analysis.

4.1 Reserved columns

Before we start, we reserved the first 19 columns as they contain the most essential information about the studies from our perspective.

```r
reserved_cols <- head(df_cols,19)
reserved_cols
```

```
##  [1] "Study Identifier"
##  [2] "Year"
##  [3] "Intervention"
##  [4] "Intervention types"
##  [5] "Intervention subgroups"
##  [6] "Country"
##  [7] "Setting"
##  [8] "Types of disease_clean"
##  [9] "Type of disease in two categories (dementia and MCI)"
## [10] "Updated Dementia Severity"
## [11] "Is this a study focusing on carers of people living with dementia or mild cognitive impairment
## [12] "Primary Cognition Measure standardized z-score change"
## [13] "Type"
## [14] "Baseline Score - Primary Cognition"
## [15] "Number of Participants at Baseline"
## [16] "Baseline SD"
## [17] "Post Treatment Score"
## [18] "Post Treatment N"
## [19] "Post Treatment SD"
```

4.2 Keyword screening

We note that there are quite a lot of depression/caregiver burden related metrics also in the columns. So we output the colnames to a seperate file named colnames.txt for manual screening of useful columns. According to our preliminary screening, we notice a number of metrics to measure caregiver burden.

4.2.1 Direct measure of care giver burdern

First we screen the column names with the keywords for direct measurement of caregiver burden

- **Hinton 2020** and **Uyar 2019:** Zarit Burden Interview (ZBI)
- **Govindakumari 2020:** Care giver burden, unspecified

```r
caregiver_burden_keys <- unique(c(
    grep("burden", df_cols),
    grep("Burden", df_cols),
    grep("ZBI", df_cols),
    grep("Zarit", df_cols)
    ))
caregiver_burden_keys <- df_cols[caregiver_burden_keys]

# show the number of selected columns
length(caregiver_burden_keys)
```

```
## [1] 91
```

There are a total of 91 columns selected. We will look at all the columns selected but for brevity of this article, we omit the output of the following code.

```r
caregiver_burden_keys
```

4.2.2 Neuropsychiatric symptom scales

Similar procedures are used to screen the colunms related to neuropsychiatric symptoms of caregivers.

Neuropsychiatric symptoms of caregivers:

- **Hinton 2020:** Depression/anxiety symptoms, measured with the Patient Health Questionnaire-4 (PHQ-4)
- **Uyar 2019:** Depression symptoms, measured with Beck Depression Inventory (BDI)
- **Uyar 2019:** Anxiety symptoms, measured with Beck Anxiety Inventory (BAI)
- **Uyar 2019:** Neuropsychiatric symptoms, measured with Neuropsychiatric Inventory-Distress (NPI-D)
- **Ghaffari 2019:** Neuropsychiatric symptoms, measured with General Health Questionnaire (GHQ-28)
- **Zarepour 2020:** Anxiety symtoms, measured with the Spielberger questionnaires
- **Pan 2019:** Depression symptoms, measured with 10-item CES-D
- **Uyar 2019:** Quality of life, measured with Quality of Life Scale SF36 (SF36 mental health)

Neuropsychiatric symptoms of patients:

- **Chen 2020** and **Pan 2019:** Mini Mental State Examinination (MMSE)
- **Uyar 2019:** Neuropsychiatric symptoms, measured with Neuropsychiatric Inventory–Severity (NPI-S)

```
# keywords for neuropsychiatric symptoms:
mental_health_keys <- unique(c(

    # abbreviations
    grep("PHQ", df_cols),
    grep("SF36", df_cols),
    grep("NPI", df_cols),
    grep("GHQ", df_cols),
    grep("PHQ", df_cols),
    grep("BDI", df_cols),
    grep("BAI", df_cols),
    # grep("MMSE", df_cols), # we don't include as this is mostly for care recipients

    # words and phrases
    grep("Neuropsychiatric", df_cols),
    grep("neuropsychiatric", df_cols),
    grep("Depression", df_cols),
    grep("depression", df_cols),
    grep("Anxiety", df_cols),
    grep("anxiety", df_cols)
    ))

# select mental health keys and drop columns and rows with only NAs
mental_health_keys <- df_cols[mental_health_keys]

# show the number of selected columns
length(mental_health_keys)
```

```
## [1] 139
```

4.2.3 Behavioural metrics

Behavioural metrics of caregivers:

- **Uyar 2019:** Quality of life, measured with Quality of Life Scale SF36 (SF36 physical health)
- **Pan 2019:** Positive/Negative coping, measured with the Simplified Coping Scale
- **Pan 2019:** Mutuality, measured with 15-item Mutuality Scale

Behavioural metrics of patients:

- **Uyar 2019:** Quality of life, measured with Quality of life in Alzheimer's Disease (QoL-AD)

- **Chen 2020:** Barthel activities of daily living scale (BADL)
- **Chen 2020:** Behavioral pathology assessment scale of Alzheimer's disease (BEHAVE-AD)
- **Govindakumari 2020:** Quality of life, unspecified
- **Pan 2019:** Quality of life, measured with the 14-item Activities of Daily Living scale (ADL)

```
# keywords for behavioural metrics:
behvaioural_keys <- unique(c(

    # abbreviations
    grep("SF36", df_cols),
    grep("QoL", df_cols),
    grep("BADL", df_cols),
    grep("BEHAVE", df_cols),
    grep("ADL", df_cols),

    # words and phrases
    grep("Quality", df_cols),
    grep("quality", df_cols),
    grep("Coping", df_cols),
    grep("coping", df_cols),
    grep("Mutuality", df_cols),
    grep("mutuality", df_cols),
    grep("daily", df_cols),
    grep("Daily", df_cols),
    grep("Barthel", df_cols),
    grep("Behaviour", df_cols),
    grep("Behaviour", df_cols),
    grep("behavior", df_cols),
    grep("behavior", df_cols)
    ))

# select behavioural keys and drop columns and rows with only NAs
behvaioural_keys <- df_cols[behvaioural_keys]
# behvaioural_keys[-grepl('sleep', behvaioural_keys)]

# show the number of selected columns
length(behvaioural_keys)
```

```
## [1] 336
```

4.3 Subsetting the dataset

We first simplify the selected studies from a data frame to a list of study identifier.

```
selected_studies <- read.csv('selected_studies.txt')
selected_studies <- unique(selected_studies$Study.Identifier)
length(selected_studies)
```

```
## [1] 27
```

Then we can subset the dataset according to our selection of studies and columns. For example, for the study that directly measures caregiver burden, we can use the following code:

```
# subset by caregiver burden-related columns
df_caregiver_burden <-
  dataset[which(rowSums(is.na(dataset[caregiver_burden_keys]))!=ncol(dataset[caregiver_burden_keys])), ]
# subset by selected studies
```

```r
df_caregiver_burden <-
  df_caregiver_burden[df_caregiver_burden$'Study Identifier' %in% as.list(selected_studies), ]

# drop rows and columns with only NA
df_caregiver_burden <-
  df_caregiver_burden[rowSums(is.na(df_caregiver_burden)) != ncol(df_caregiver_burden), ]
df_caregiver_burden <-
  df_caregiver_burden[,colSums(is.na(df_caregiver_burden))<nrow(df_caregiver_burden)]

# show the size of simplified dataset
dim(df_caregiver_burden)
```

```
## [1]  22 164
```

Thus, the dataset is significantly reduced, which we can manually curate for a meta analysis-ready dataset. Similary, we can generate the dataset for caregiver mental health and caregiver quality of life (behaviour):

```r
# subset for caregiver mental health
df_mental_health <-
  dataset[which(rowSums(is.na(dataset[mental_health_keys]))!=ncol(dataset[mental_health_keys])), ]
df_mental_health <-
  df_mental_health[df_mental_health$'Study Identifier' %in% as.list(selected_studies), ]
df_mental_health <-
  df_mental_health[rowSums(is.na(df_mental_health)) != ncol(df_mental_health), ]
df_mental_health <-
  df_mental_health[,colSums(is.na(df_mental_health))<nrow(df_mental_health)]

# print dimensions of the dataset
dim(df_mental_health)
```

```
## [1]  4 87
```

```r
# subset for caregiver behaviour or quality of life
df_behavioural <-
  dataset[which(rowSums(is.na(dataset[behvaioural_keys]))!=ncol(dataset[behvaioural_keys])), ]
df_behavioural <-
  df_behavioural[df_behavioural$'Study Identifier' %in% as.list(selected_studies), ]
df_behavioural <-
  df_behavioural[rowSums(is.na(df_behavioural)) != ncol(df_behavioural), ]
df_behavioural <-
  df_behavioural[,colSums(is.na(df_behavioural))<nrow(df_behavioural)]

# print dimensions of the dataset
dim(df_behavioural)
```

```
## [1]  16 163
```

Eventually, we can write these subsets to a single Excel file for further curation, with the following code:

```r
# create the workbook
wb <- createWorkbook()

# add the worksheets
addWorksheet(wb, "caregiver_Burden")
addWorksheet(wb, "caregiver_MentalHealth")
addWorksheet(wb, "caregiver_Behaviour")
```

```r
# write the workbook
writeData(wb, "caregiver_Burden", df_caregiver_burden)
writeData(wb, "caregiver_MentalHealth", df_mental_health)
writeData(wb, "caregiver_Behaviour", df_behavioural)

# save the workbook
saveWorkbook(wb, "Dataset_data_filtered_final.xlsx", overwrite = TRUE)
```